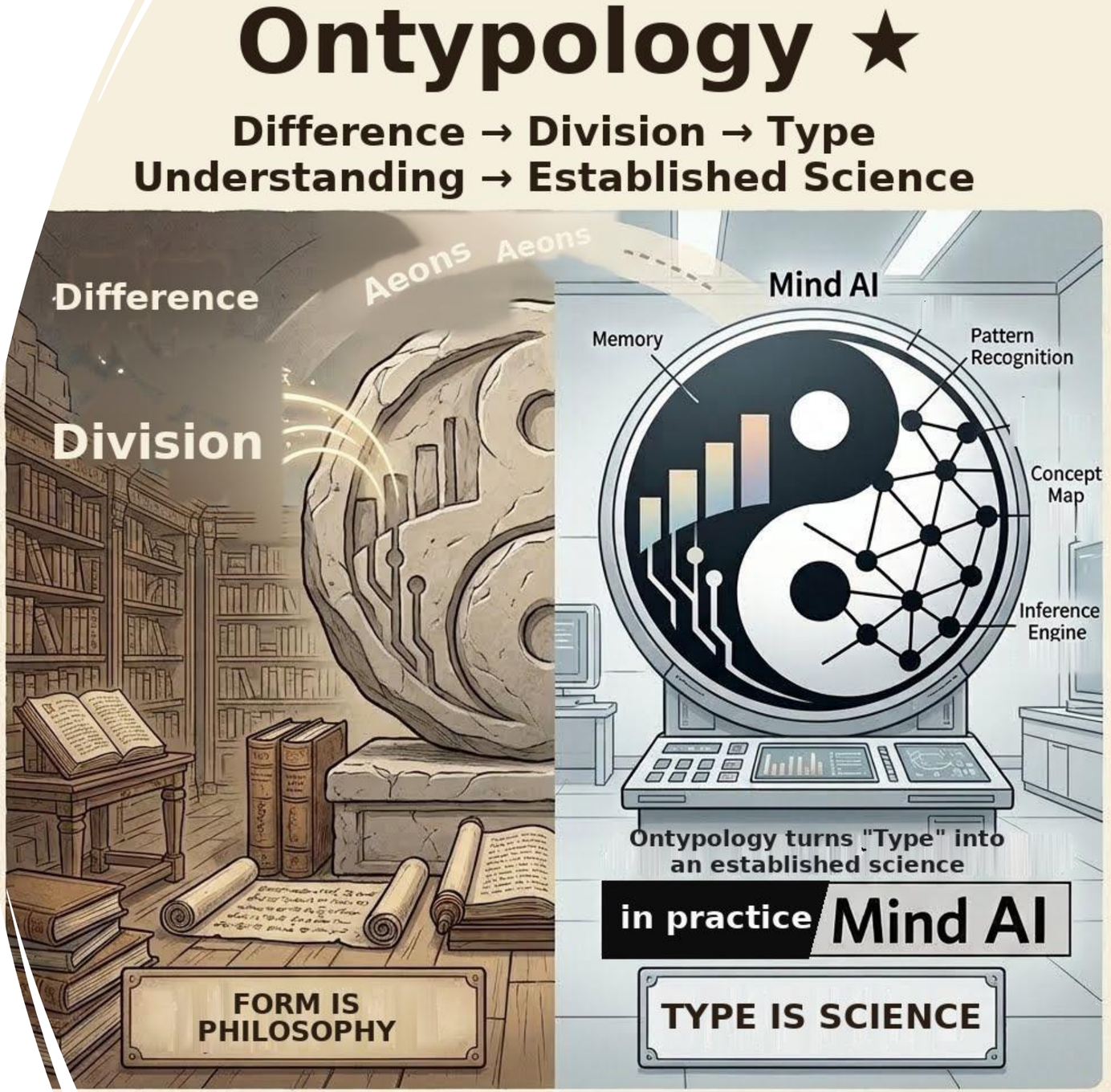
Ontypology

- From difference to division, from division to type, from type to understanding – the brain has walked this path for aeons.
- A century ago, logic was philosophy; once seen clearly, it became an established science.
- Today we explore "Type" (a brain concept) as philosophy; Ontypology makes "Type" clear, turns it into an established science, and puts it into practice as Mind AI.





AI Bio Lab Brain

From loose data to a reasoning-capable replica: every experiment certain, acceptable, traceable

Full architecture analysis

S1–S7 onboarding

Modular dashboard

Turning 'certainty' into a method.

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The collapse of the traditional bench



Inconsistent definitions

Same data, two people
The interpretation differs.

The process depends on handover

The SOP lives in the veteran's hands,
Handover and it walks away.

Data without traceback

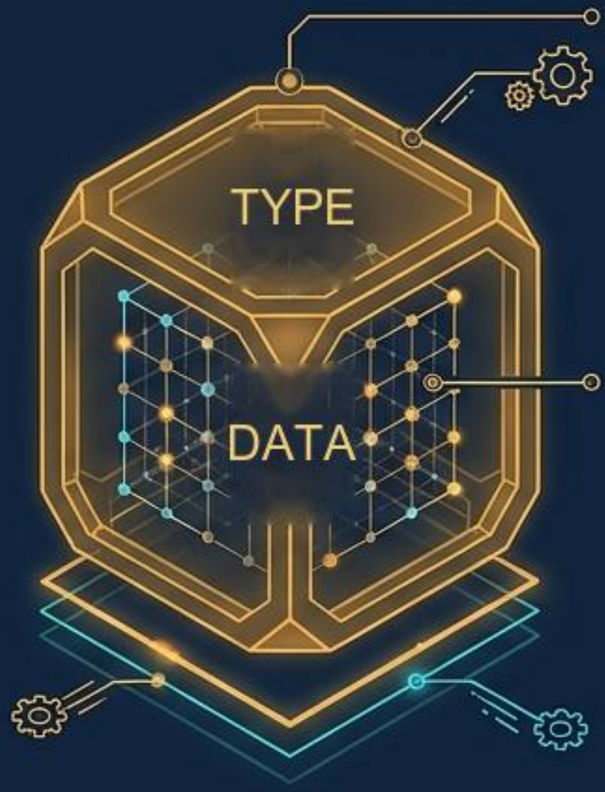
Abbreviation: read to the end only

"At first it seemed…

Specification by feeling

Batch comparison only "against the previous batch "roughly" has no number.

Reasoning-capable experiment replica



Type: provides the reasoning skeleton (replica container · recipe group (generation · fault category)



Data: provides the measurement flesh (concentration, initial rate, statistics)



What a lab truly preserves is not its instruments but the collective memory of "what we did, what we saw, and why this conclusion"

New person, six months later: the replica still runs.

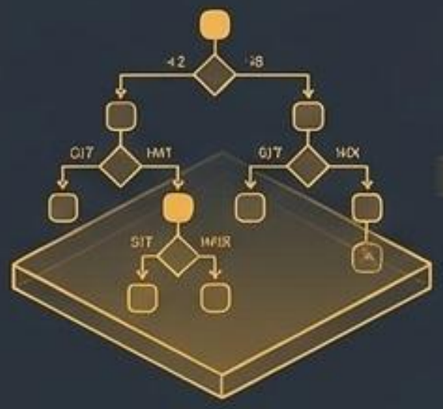
MIND AI

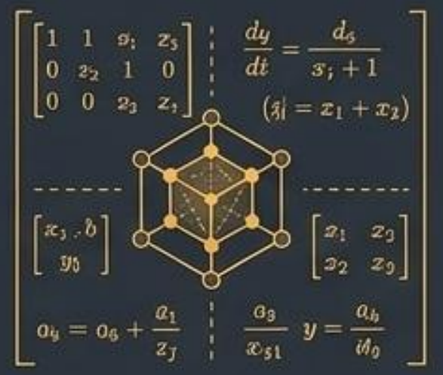
LLM engine – language



Language understanding and generation, induction and statistics

Mind Model engine – wisdom





Deductive reasoning · Type structure management · long-term memory

Lab placement

· literature review · first-draft output · data cleaning · report polishing

Lab placement

· six-Type modelling · skeleton derivation · n(T) acceptance and evolution

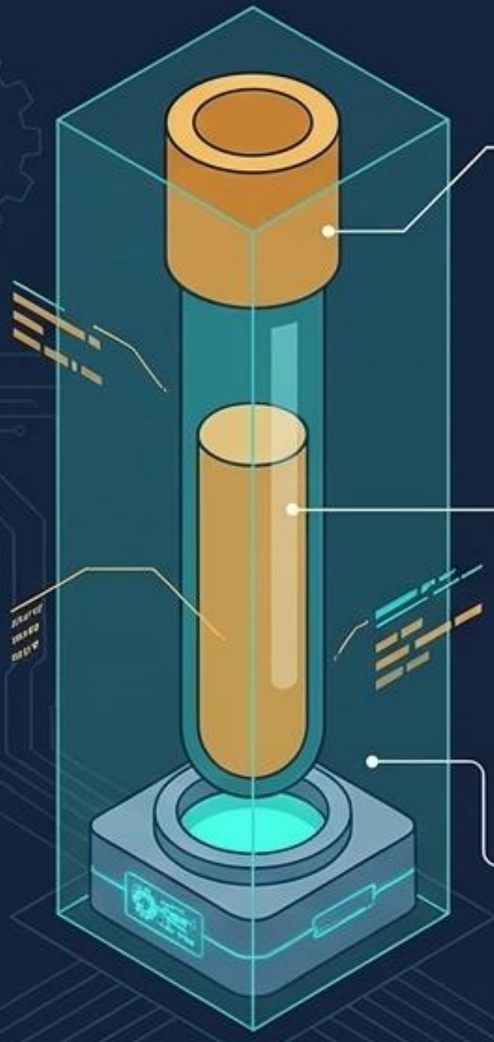


Referee ladder: n(T) verdict

> The LLM talks fast, the Mind Model talks true; Language to the LLM, wisdom to the Mind Model, and finally the n(T) criterion as referee

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Space Type: what it is



( ) Container: C = (id(C), S(C))

Well identity unchanged (reagent changed

(still well A1) ∘

{ } Composition BOM. Reaction recipe vs. remove-check pass rate

(remove the substrate and the whole loses identity

(loss) ∘

[ ] Abstract: T = (F(T), n(T))

Experiment category and pass criterion (each

(only when the features pass).

Time Type: how to do it





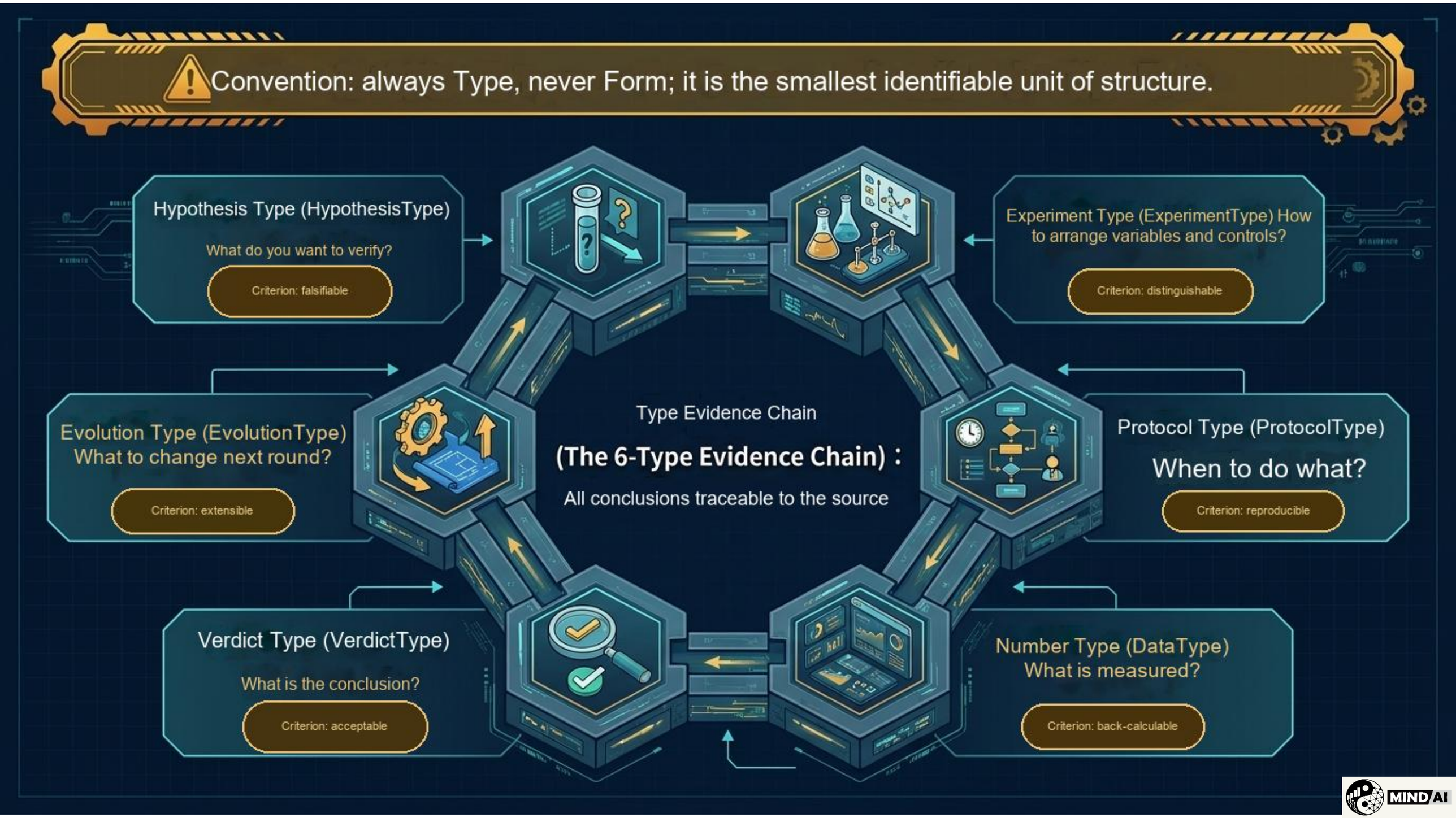


Verb = function = method = move

- Search group (literature review)
- Generation group (apply the design)
- Thinking group (data analysis)
- Evolution group (batch learning)

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Convention: always Type, never Form; it is the smallest identifiable unit of structure.



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BIOTECH BLUEPRINT

Hypothesis Type card – Hypothesis Card

Objective: understand this enzyme batch

Falsifiable condition: Km and Vmax of the new rALP batch

Falls within the previous batch's 95% confidence interval

Variable Tripartite Design Console

Independent variable: pNPP concentration

6 gradients 0.05→1.60 mM

Dependent variable: Initial rate v

(ΔA405/min，0–90 s linear segment)

0.000 0.150 0.300 0.450

Control variables: temperature 25 °C, pH 9.8 locked

TEMP : 25°C

pH : 9.8

Control group: blank + heat-denatured enzyme + previous-batch standard (all three required)

blank

Heat-denatured enzyme

Previous-batch standard

None may be missing

Remove-check radar

Composition

Essential part

Enzyme · substrate

Remove it and identity is lost; what is measured is not this

Degraded part

Buffer, stop solution

Function degraded

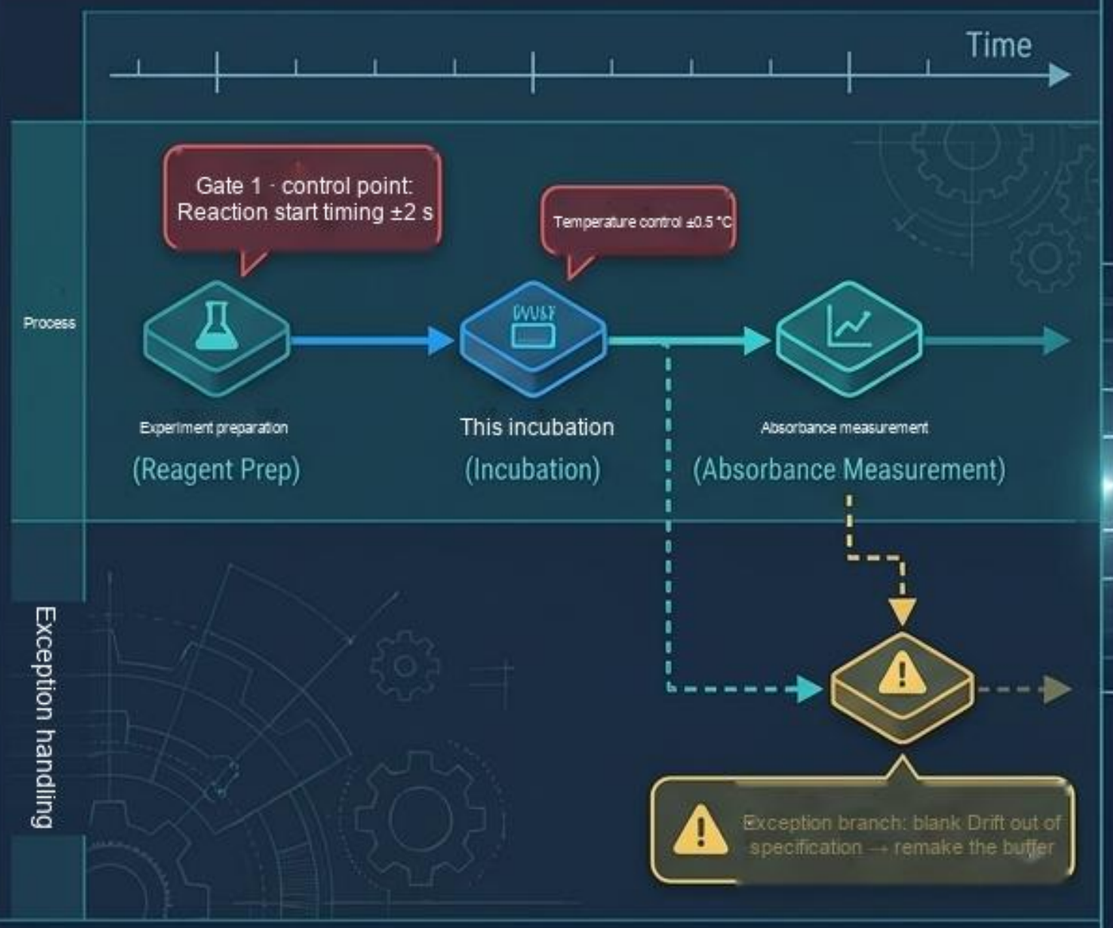
Non-essential part

Label dye

No difference if removed

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Protocol Type (SOP-ALP-042v2.1)



Reproducible by others

Number Type (DataType)



Traceability triad

Tag 1
Instrument ID
(UV-2600#0412)

Tag 2
Calibration record
(Calibration Record)

Tag 3
Operator
(Engineer Wu)

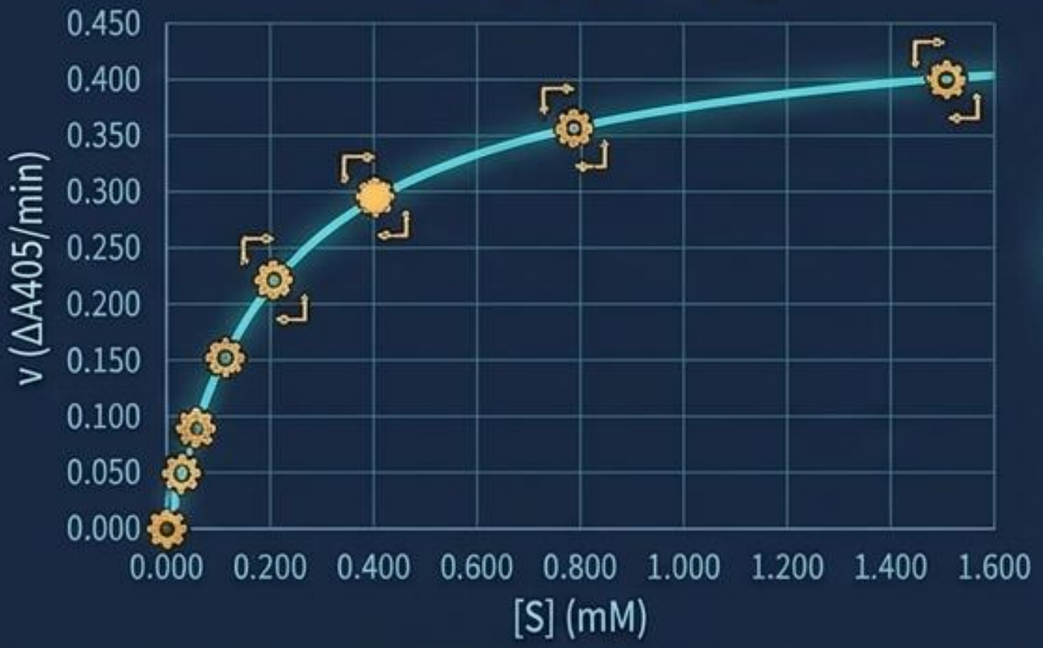
What fails the n(T) card is not merely worse; it is "not permitted for use"

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Verdict Type (VerdictType)

Method: skeleton first, numbers after

$$v = V_{max}[S]/(K_m+[S])$$



Required output: parameter estimates (Km = 0.25 mM) +

Uncertainty (95% CI) + applicable range (0.05–1.60 mM)

Evolution Type (EvolutionType)

Lessons recorded

Blank drift → Buffer to be prepared fresh on the day

v2.2

Next-round hypothesis

at 25/30/37 °C Km is temperature dependent

SOP v2.1

Versions have a story: a lab without an evolution log, ten years of work

A thousand experiments are really one experiment repeated a thousand times.

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Layer 4: Type evidence chain

The conclusion traces back along a luminous path to the hypothesis; all six stations are checkable.

Third layer: Empirical Validation

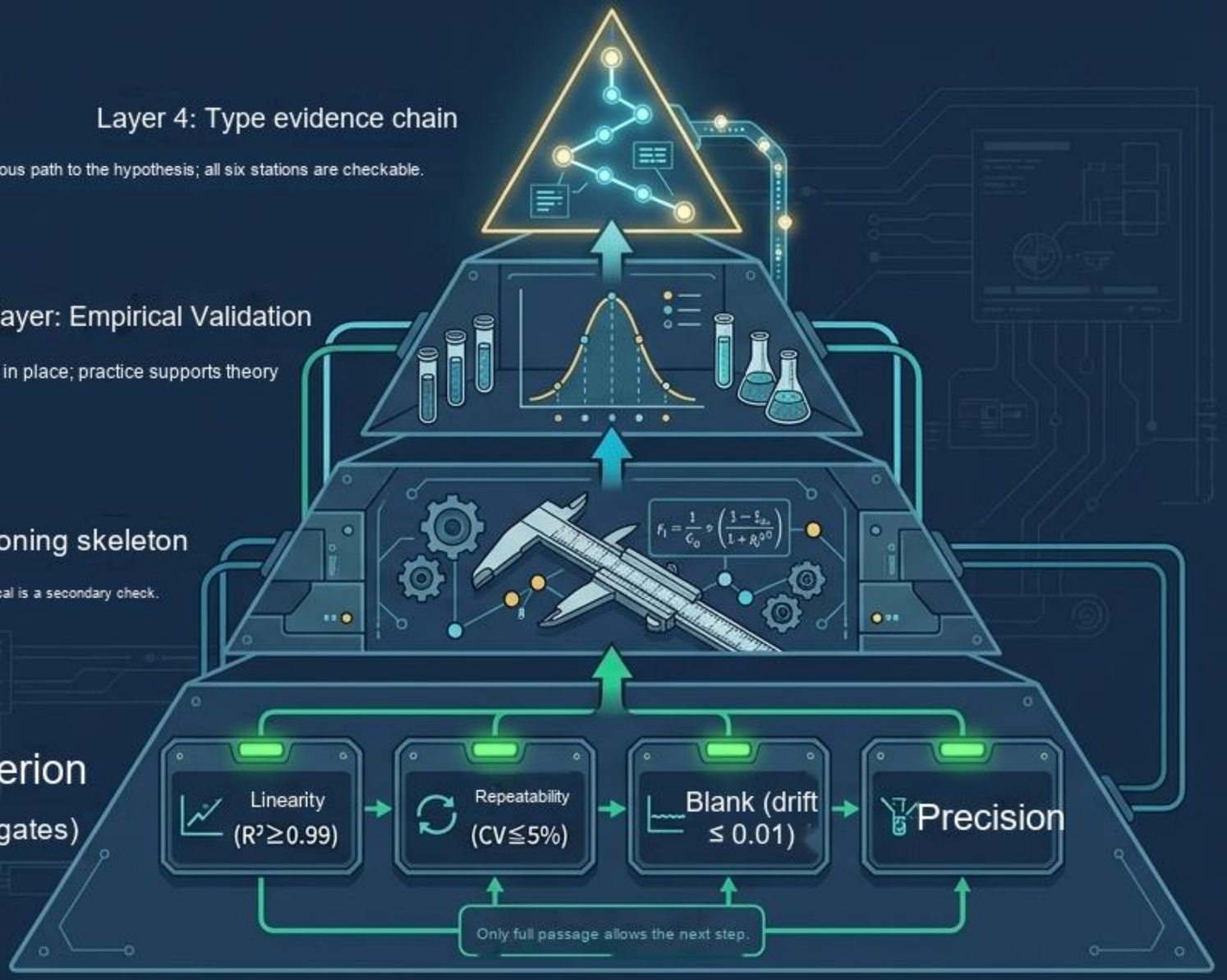
Controls, replicates and statistics all in place; practice supports theory

Layer 2: reasoning skeleton

The M-M equation anchors the reasoning; the double reciprocal is a secondary check.

Layer 1: n(T) criterion

(four data-quality gates)



Core reminder: certainty cannot be guaranteed by feeling; it needs a strict card.

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Top left (match /):

Type built right, process still skips

➡SOP training, then rerun

Process stage.



Top right (match / yes) :

Correct conclusion, reproducible process

→ Report release approved, entered into the Type

Evidence chain.



Bottom left (/): Total breakdown

➡Output archived for teaching

Return to S1 and start over.



Bottom right (/ yes):

Process smooth, but Type built wrong

➡Return to axis-by-axis Type building and redefine

Meaning, structure and verdict.



Y-Axis

錯

錯

對

X-Axis

Time Type (does the process run?)

Space verifies structure, time verifies process; only both axes correct counts as certain

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Rule 1: one stage at a time; skipping a stage = skipping a gate
S1: Requirements and hypothesis
→ [Output: hypothesis Type card]
S2 Experiment design
[Output: experiment design doc]
S3: Protocol authoring
[Output: SOP process]
S4: Data design
[Output: data table spec]
S5: Interpretation and analysis
[Output: verdict report]
S6: Report consolidation
[Output: integrated report]
S7: Acceptance and evolution
→ [Output: evolution log]
S2: Experiment design
[Output: experiment design doc]
S3: Protocol authoring
[Output: SOP process]
S4: Data design
[Output: data table spec]
S5: Interpretation and analysis
→ [Output: verdict report]
S6: Report consolidation
→ [Output: integrated report]

Engineer Wu's ALP kinetics assay case (rALP new-batch equivalence comparison

A: S1+S2 setup

Objective:

New batch Km/Vmax Falls within the previous batch's 95% CI.

Independent-variable design matrix:

pNPP 6-concentration gradient

(0.05-1.60 mM) ∘

S4: Data reality

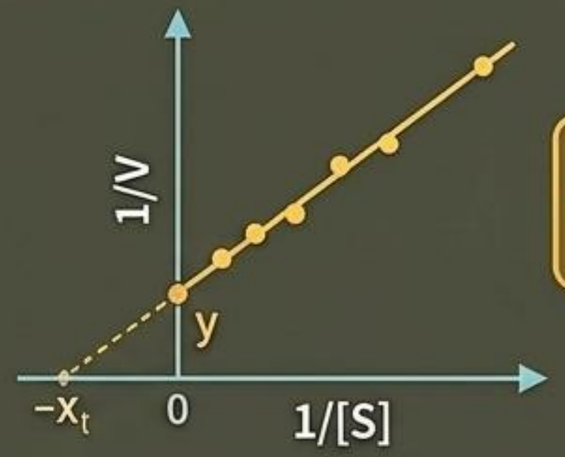
raw initial rates in triplicate (n = 3)

	S1	S2	S3	Avg:
1	0.125	0.126	0.124	0.125
2	0.125	0.123	0.125	0.125
3	0.125	0.127	0.128	0.125
4	0.126	0.128	0.121	0.124
5	0.129	0.124	0.122	0.125
6	0.119	0.125	0.124	0.125

Pass

Four gates passed:
Linearity R^2 = 0.997,
CV=1.8%,
blank=0.004...

S5 Interpret the results



Previous-batch CI overlap
Equivalence established

Result: Km = 0.25 ± 0.02 mM

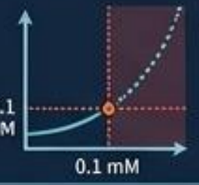
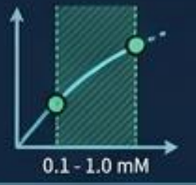
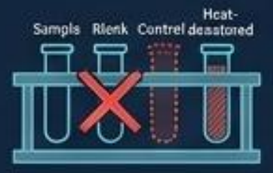
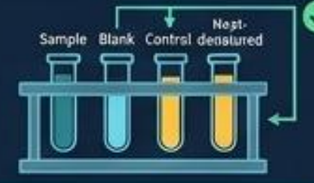
S7 evolution log

SOP v2.1 → v2.2

(into the temperature-gradient stage)

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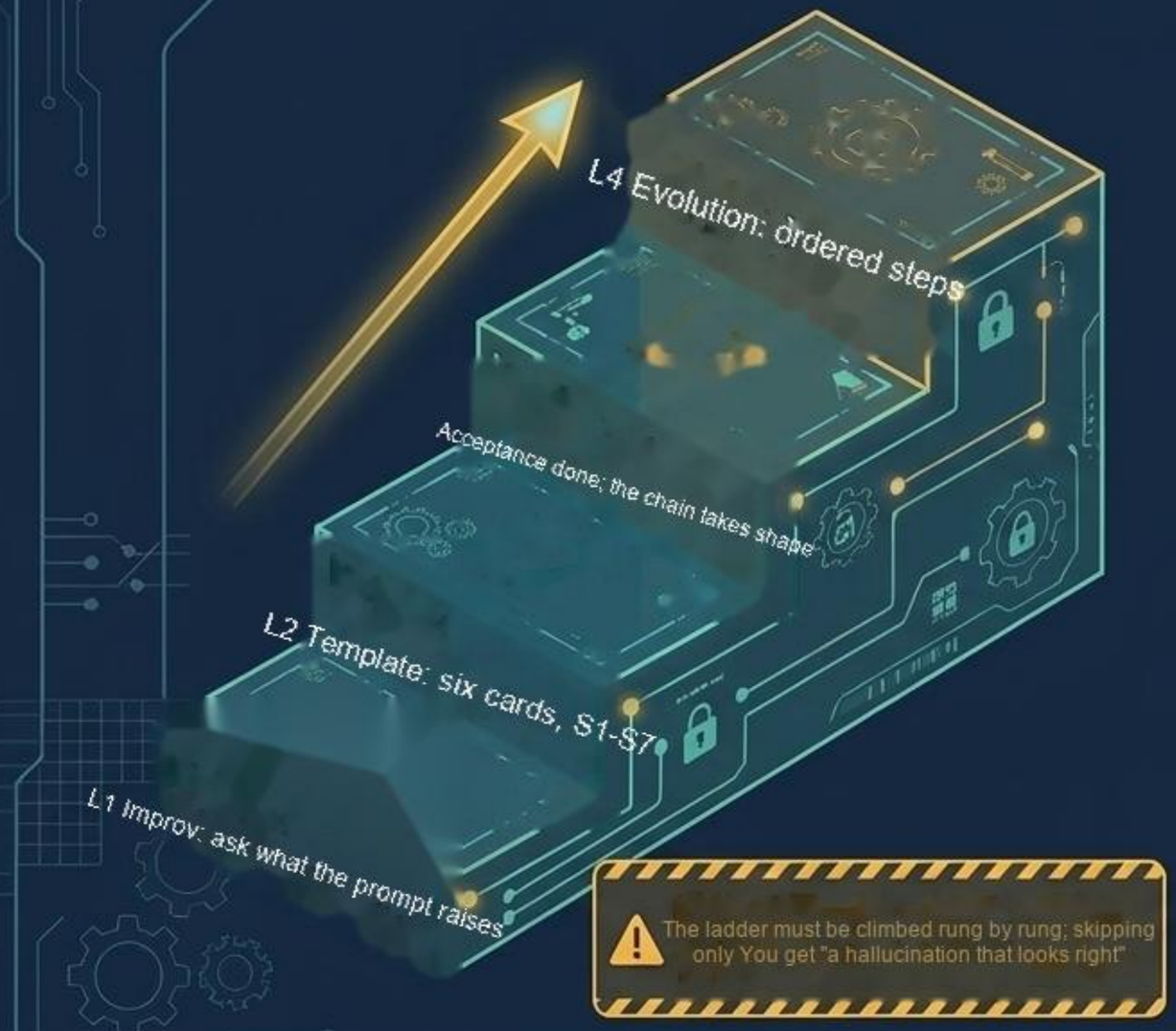
Engineer Wu's ALP kinetics assay case (rALP new-batch equivalence comparison) — data diagnosis and correction comparison table

Symptom: rising absorbance = high activity 	[Correlation treated as causation]	Correct: subtract the blank first, check linearity, and let mechanism support causation 
Fault: extrapolate from 1.6 mM to 0.01 mM 	[Extrapolation out of bounds]	Correct: state the applicable range Conclusions do not cross the boundary. 
Symptom: read the 30 min end-point value 	[End-point method passed off as initial rate]	Correct: slope of the 0–90 s linear segment is the true v. 
Fault: omit blank/linearity 	[Missing control]	Correct: the control is an essential part; it is missing then the experiment loses its identity. 
Symptom: n = 1 and a conclusion is written, no SD 	[Insufficient replication]	Correct: n ≥ 3, point estimate and 95% CI. 

Common thread: skipping the verdict, omitting controls, overstating conclusions — none of the three improves the numbers, all three endanger the conclusion

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Engineer Wu's ALP kinetics assay case



⚠ The ladder must be climbed rung by rung; skipping only You get "a hallucination that looks right"

Onboarding needs no big bang

Month 1 (pilot point)	Month 2 (layout)	Month 3 (normal state)
Pick a familiar process and run the seven stages	Six Type cards templated · rolled out to all group training	n(T) acceptance enters the review; an evolution log is emitted

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1. Experiment replica (Type + Number)

Take away an architecture where memory is not lost · structure reused
Record structure as Type, measurement as Number; change hands

Every batch-change path is preserved.



2. Six Types In one click Update (hypothesis, experiment, process, data, verdict, evolution)

Take away the S1–S7 prompt pipeline · six Types

Interlocking links, no blind spot in traceback.



3. Four-layer gatekeeping (verifiable with certainty)

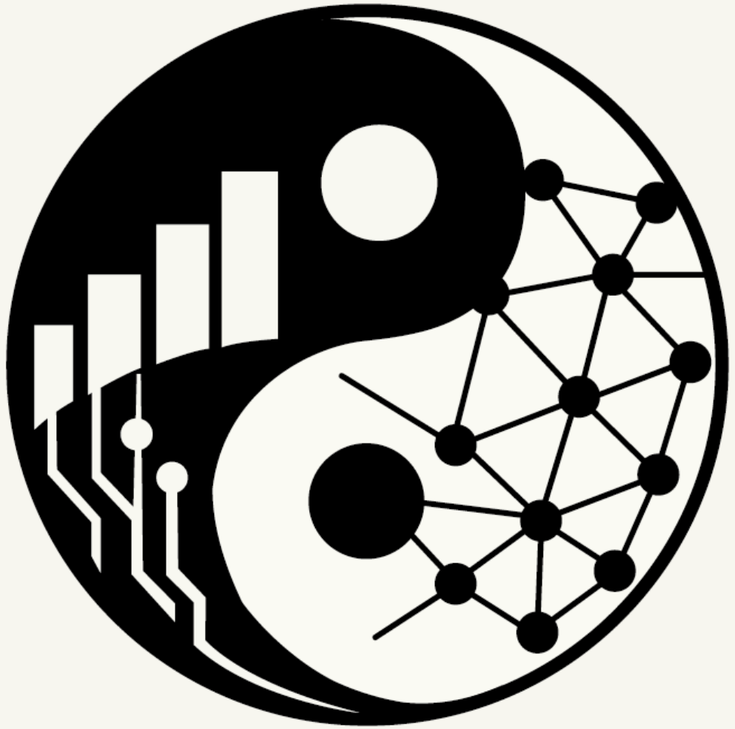
Take away the space × time cross-acceptance and n(T)

Verdict. Certainty is not luck; it is method.

Note "certainty" Adjectives become methods · Convention: always Type, never Form

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LonGreat Vision: to create intelligent life that understands the essence of the world



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"We aim to make artificial intelligence not merely a tool, but a system of intelligent life that truly possesses the abilities to understand, reason, learn, and create."

From the unification of knowledge, to the unification of thought, and finally toward the unification of intelligence – this is the eternal path to AGI that LonGreat pursues.

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